

 Marwadi University Marwadi Chandarana Group  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date: 28/08/2026	Enrollment No: 92400133030

Theory:

Histogram:

A histogram is a graphical representation that shows the distribution of pixel intensities in an image. It displays the frequency of each intensity value on the horizontal axis and the number of pixels with that intensity value on the vertical axis.

Histogram Equalization:

Histogram equalization is a technique used to enhance the contrast of an image by redistributing the intensity values. It stretches the histogram of an image to cover the full range of intensity values, resulting in a more balanced and visually appealing image.

Histogram Matching:

Histogram matching is a process of transforming an image to match the histogram of a reference image. It is used to transfer the characteristics of the reference image's histogram to the input image.

Flowchart:

The flowchart for the program will consist of the following steps:

- Load the input image.
- Generate and display the histogram of the image.
- Apply histogram equalization to enhance the contrast.
- Generate and display the histogram of the equalized image.
- Load the reference image for histogram matching.
- Perform histogram matching between the input image and reference image.
- Generate and display the histogram of the matched image.
- Display the original image, equalized image, and matched image side by side.
- Save the equalized and matched images (optional).

Program

Now, let's see the Python program that implements histogram generation, histogram equalization, and histogram matching:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
def generate_histogram(image):
    # Calculate the histogram of the image
    histogram = cv2.calcHist([image], [0], None, [256], [0, 256])
    return histogram
def perform_histogram_equalization(image):
    # Convert image to grayscale
    gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    # Apply histogram equalization
    equalized_image = cv2.equalizeHist(gray_image)
    return equalized_image
```

 Marwadi University Marwadi Chandarana Group  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:28/08/2026	Enrollment No:92400133030

```

def perform_histogram_matching(input_image, reference_image):
# Convert images to grayscale
gray_input = cv2.cvtColor(input_image, cv2.COLOR_BGR2GRAY)
gray_reference = cv2.cvtColor(reference_image, cv2.COLOR_BGR2GRAY)

# Calculate histograms
input_hist = generate_histogram(gray_input)
reference_hist = generate_histogram(gray_reference)

# Perform histogram matching
matched_image = cv2.matchHistograms(gray_input, gray_reference,
method=cv2.HISTCMP_HELLINGER)
return matched_image

# Load the input image
input_path = 'input_image.jpg'
input_image = cv2.imread(input_path)

# Generate and display the histogram of the input image
input_hist = generate_histogram(input_image)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.show()

# Perform histogram equalization
equalized_image = perform_histogram_equalization(input_image)

# Generate and display the histogram of the equalized image
equalized_hist = generate_histogram(equalized_image)
plt.plot(equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.show()

# Load the reference image for histogram matching
reference_path = 'reference_image.jpg'
reference_image = cv2.imread(reference_path)

# Perform histogram matching
matched_image = perform_histogram_matching(input_image, reference_image)

# Generate and display the histogram of the matched image
matched_hist = generate_histogram(matched_image)

```

 Marwadi University Marwadi Chandarana Group  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date: 28/08/2026	Enrollment No: 92400133030

```
plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.show()
```

```
# Display the original image, equalized image, and matched image side by side
combined_image = np.hstack((input_image, equalized_image, matched_image))
cv2.imshow("Original | Equalized | Matched", combined_image)
cv2.waitKey(0)
```

```
# Save the equalized and matched images (optional)
equalized_path = 'equalized_image.jpg'
matched_path = 'matched_image.jpg'
cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
print(f'Equalized image saved at: {equalized_path}')
print(f'Matched image saved at: {matched_path}')
```

Explanation:

We import the required modules, including cv2 for image processing, numpy for array operations, and matplotlib.pyplot for plotting the histogram.

The generate_histogram function takes an image as input and calculates its histogram using the cv2.calcHist function.

The perform_histogram_equalization function converts the image to grayscale, applies histogram equalization using cv2.equalizeHist, and returns the equalized image.

The perform_histogram_matching function performs histogram matching between the input image and the reference image using cv2.matchHist with the cv2.HISTCMP_HELLINGER method.

In the main part of the program, we load the input image using cv2.imread.

We generate and display the histogram of the input image using the generate_histogram function and matplotlib.pyplot.plot.

Histogram equalization is performed using the perform_histogram_equalization function, and the equalized image is stored in the equalized_image variable.

We generate and display the histogram of the equalized image using the same process as step 6.

The reference image is loaded for histogram matching.

Histogram matching is performed using the perform_histogram_matching function, and the matched image is stored in the matched_image variable.

We generate and display the histogram of the matched image using the same process as step 6.

The original image, equalized image, and matched image are displayed side by side using

 Marwadi University Marwadi Chandarana Group  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date: 28/08/2026	Enrollment No: 92400133030

cv2.imshow and np.hstack.

The equalized and matched images are saved to files using cv2.imwrite.

You can replace the input_path and reference_path variables with the paths to your input and reference images, respectively.

Remember to have OpenCV and matplotlib installed (pip install opencv-python matplotlib) before running this program.

Expected Output:

Histogram plots of the input image, equalized image, and matched image will be displayed in separate windows.

A combined image window will appear showing the original image, equalized image, and matched image side by side.

The equalized image and matched image will be saved to separate files, and their paths will be printed as output.

Note: The program assumes that the input image and reference image are in JPEG format, but you can modify the code to support other formats if needed.

Histogram Equalization:

```
import cv2
import matplotlib.pyplot as plt

# Load the image
image_path = '/Users/Desktop/SEM5/DSIP_PRACT/Experiments/Experiment_8/image.png'
# Replace with the path to your image
image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

# Calculate the histogram
histogram = cv2.calcHist([image], [0], None, [256], [0, 256])

# Plot the histogram
plt.figure(figsize=(8, 6))
plt.title('Histogram')
plt.xlabel('Pixel Value')
```

 Marwadi University Marwadi Chandarana Group NAAC  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:28/08/2026	Enrollment No:92400133030

```

plt.ylabel('Frequency')
plt.plot(histogram)
plt.xlim([0, 256])
plt.grid(True)
plt.show()

# Perform histogram equalization
equalized_image = cv2.equalizeHist(image)

# Display the original and equalized images
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.title('Original Image')
plt.imshow(image, cmap='gray')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.title('Equalized Image')
plt.imshow(equalized_image, cmap='gray')
plt.axis('off')

# Calculate the histogram
histogram = cv2.calcHist([equalized_image], [0], None, [256], [0, 256])

# Plot the histogram
plt.figure(figsize=(8, 6))
plt.title('Histogram')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')
plt.plot(histogram)
plt.xlim([0, 256])
plt.grid(True)
plt.show()

plt.tight_layout()
plt.show()

```



University
Engineering & Technology
Department of Information and Communication Technology

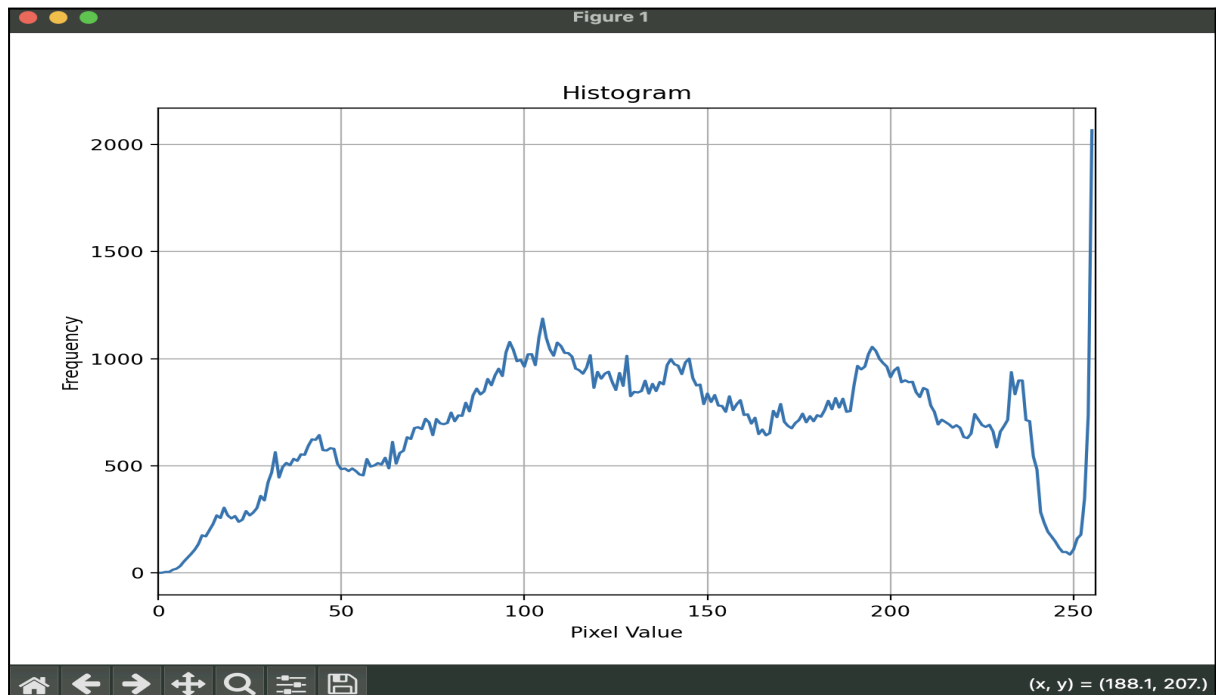
**Subject: Digital Signal and
Image Processing (01CT1513)**

**AIM: Generate Histogram of images, apply Histogram equalization
and Histogram matching on it.**

Experiment No: 8

Date: 28/08/2026

**Enrollment
No: 92400133030**



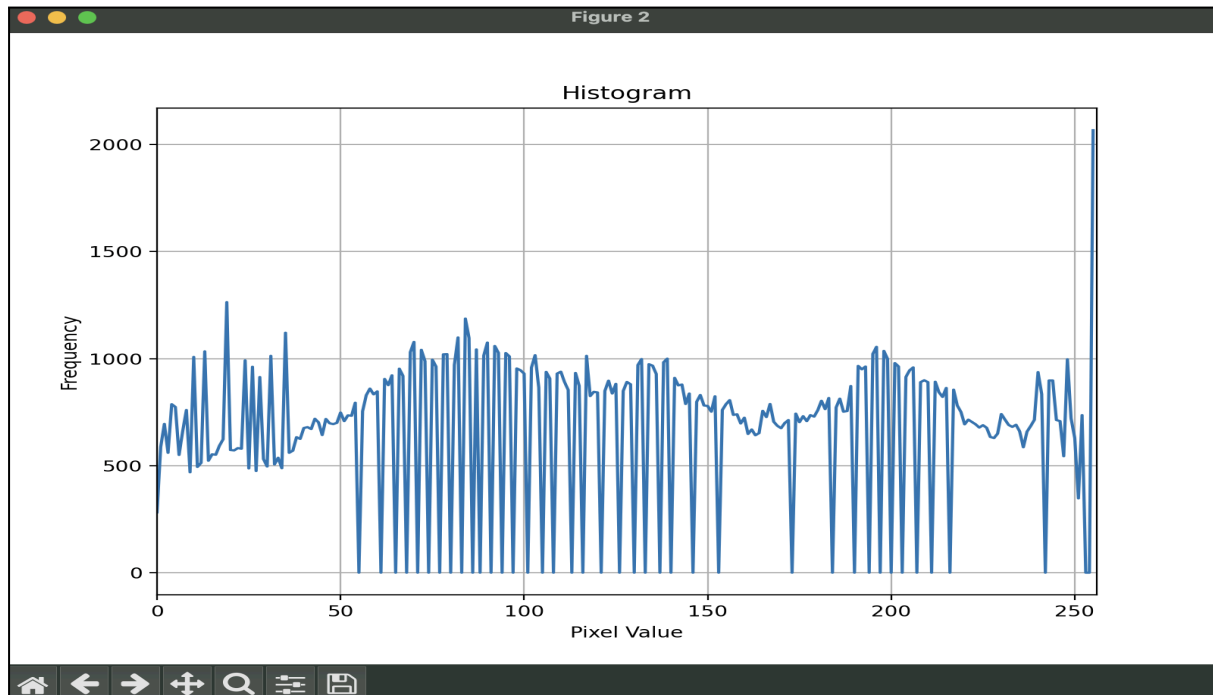
Subject: Digital Signal and Image Processing (01CT1513)

AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.

Experiment No: 8

Date: 28/08/2026

Enrollment
No: 92400133030



Histogram Matching:

 Marwadi University Marwadi Chandarana Group  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:28/08/2026	Enrollment No:92400133030

```

import cv2
import numpy as np
from matplotlib import pyplot as plt

# Load the source and reference images
source_path = '/Users/Desktop/SEM5/DSIP_PRACT/Experiments/Experiment_8/image.png'
reference_path = '/Users/Desktop/SEM5/DSIP_PRACT/Experiments/Experiment_8/Reference.png'
source_image = cv2.imread(source_path, cv2.IMREAD_GRAYSCALE)
reference_image = cv2.imread(reference_path, cv2.IMREAD_GRAYSCALE)

# Calculate histograms for the source and reference images
source_hist = cv2.calcHist([source_image], [0], None, [256], [0, 256])
reference_hist = cv2.calcHist([reference_image], [0], None, [256], [0, 256])

# Normalize histograms to have sum equal to 1
source_hist /= source_hist.sum()
reference_hist /= reference_hist.sum()

# Calculate cumulative distribution functions (CDF) for histograms
source_cdf = source_hist.cumsum()
reference_cdf = reference_hist.cumsum()

# Perform histogram matching by mapping source CDF to reference CDF
mapping = np.interp(source_cdf, reference_cdf, range(256))
matched_image = mapping[source_image]

# Convert to uint8 data type
matched_image = matched_image.astype(np.uint8)

# Display the images using Matplotlib
plt.figure(figsize=(12, 6))

plt.subplot(131)
plt.title('Source Image')
plt.imshow(source_image, cmap='gray')
plt.axis('off')

```


 Marwadi University Marwadi Chandarana Group  University Engineering & Technology Department of Information and Communication Technology		
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date: 28/08/2026	Enrollment No: 92400133030

```
plt.subplot(132)
plt.title('Reference Image')
plt.imshow(reference_image, cmap='gray')
plt.axis('off')

plt.subplot(133)
plt.title('Matched Image')
plt.imshow(matched_image, cmap='gray')
plt.axis('off')

plt.tight_layout()

plt.show()
```



GITHUB LINK:-